

MEDASSISTAI RESEARCH

SYSTEM ANALYSIS & DESIGN

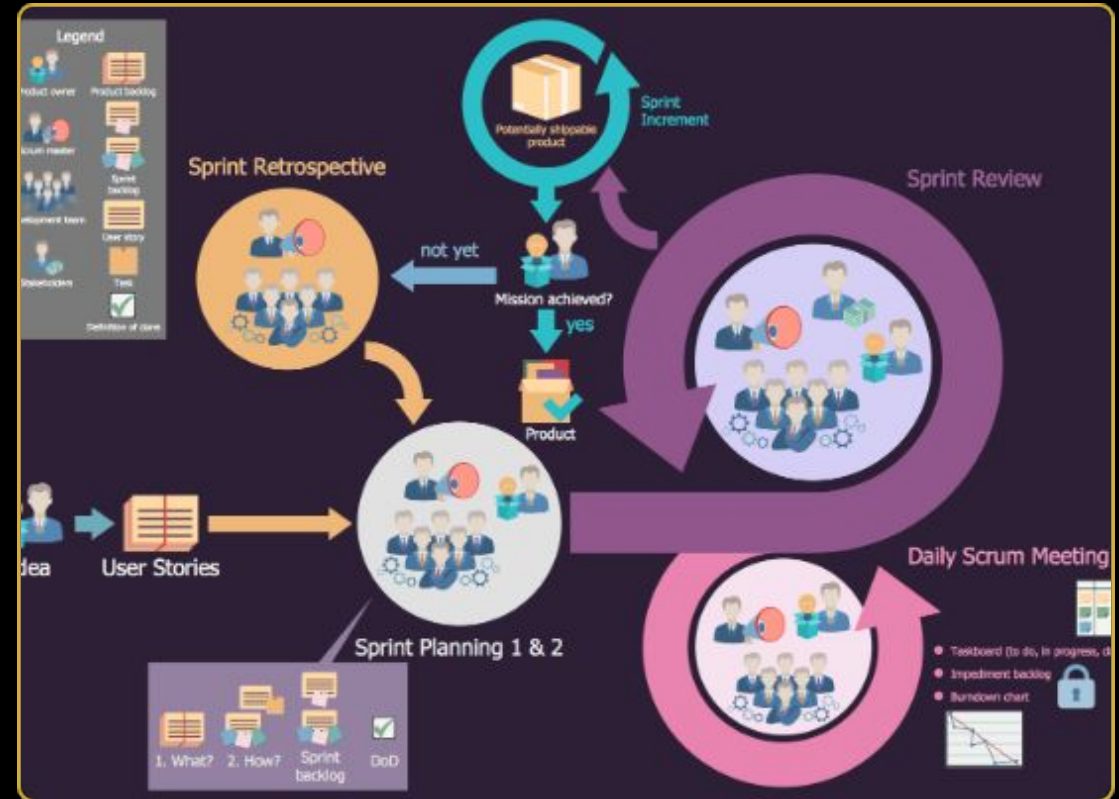
Chapter 3: Defining the Blueprint for AI-Driven
Medication Adherence in Kenya

| System Development Methodology

Design Science Research (DSR)

Guided by the DSR framework, MedAssistAI focuses on building and testing a novel innovation that solves the critical problem of non-adherence through artifact creation.

- ✔ **Agile-Scrum Hybrid:** Two-week sprints for rapid iteration.
- ✔ **Stakeholder Feedback:** Constant validation from Kenyan healthcare providers.



| Feasibility Assessment



Economic

Rating: High. Leveraging open-source stacks and free-tier APIs keeps KES costs between 25k–45k.



Technical

Rating: Medium. Ambition of Fadhili AI balanced by reliable Gemini fallback and scalable Node.js/PostgreSQL backend.



Legal

Rating: High. Designed for **Kenya Data Protection Act (2019)** compliance from inception.

| Key Insights from Data Collection

80%

Of Kenyan chronic patients miss medications at least monthly.

75%

Prefer Swahili or Bilingual support over English-only interfaces.



Source: Stakeholder interviews (7 Providers) and Patient Questionnaires (20 Patients).

| Functional Requirements (FR)

User Mgmt

FR1.1: Registration with phone, name, and chronic conditions.

FR1.4: 6-Digit PIN authentication for secure access.

AI Logic

FR3.1: Fadhili AI as primary personalized reminder engine.

FR3.3: Auto-failover to Gemini API if primary engine fails.

Notifications

FR4.1: In-app reminders for active users.

FR4.2: SMS fallback for users without internet connectivity.

| Use Case Analysis

The system interacts with three primary actors: **Patients**, **Caregivers**, and **Healthcare Providers**.

- ✓ **Patient:** Manages meds, receives AI tips, logs adherence.
- ✓ **Caregiver:** Monitors patient remotely.
- ✓ **Provider:** Views adherence dashboards for clinical follow-ups.



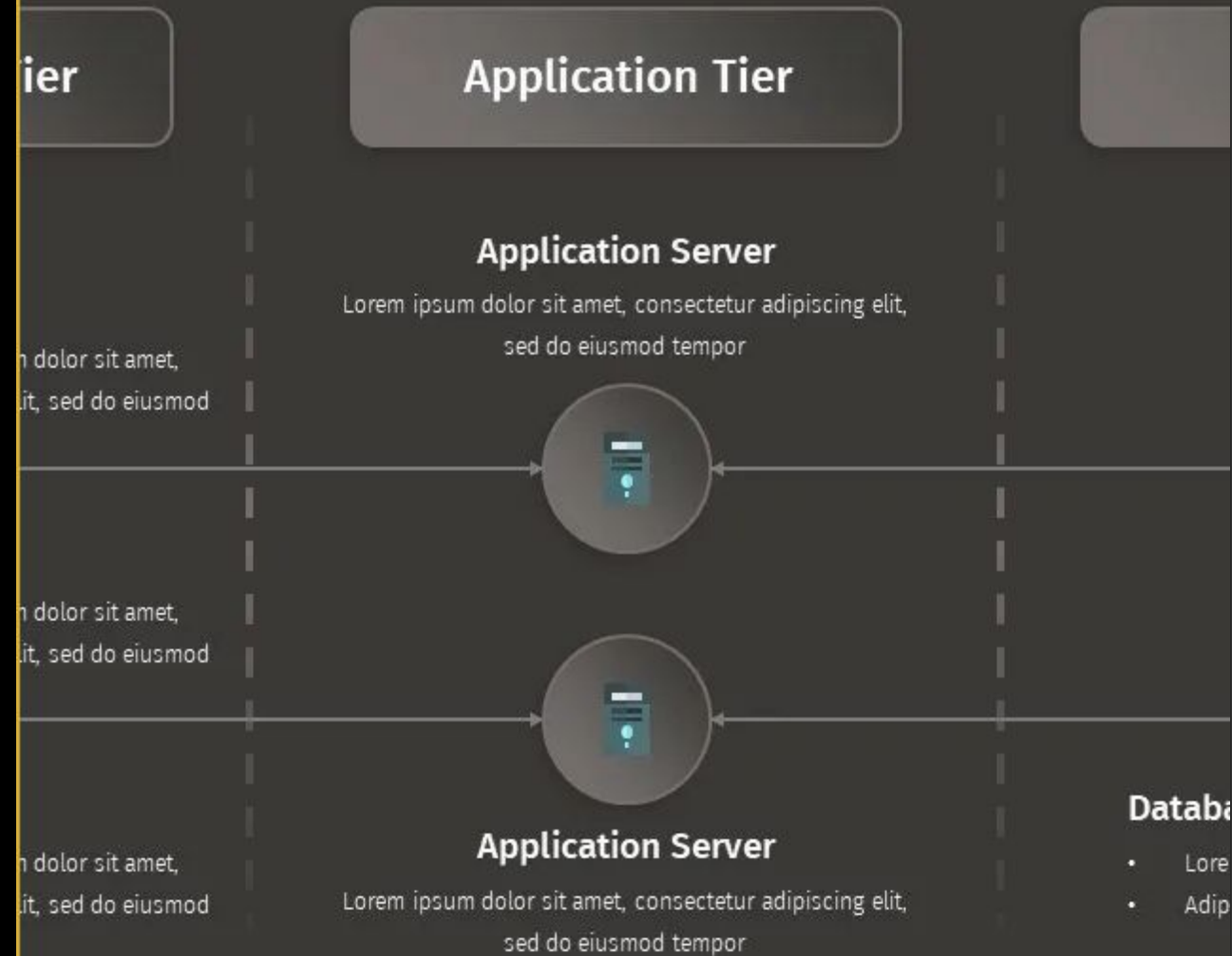
| System Architecture

The Three-Tier Blueprint

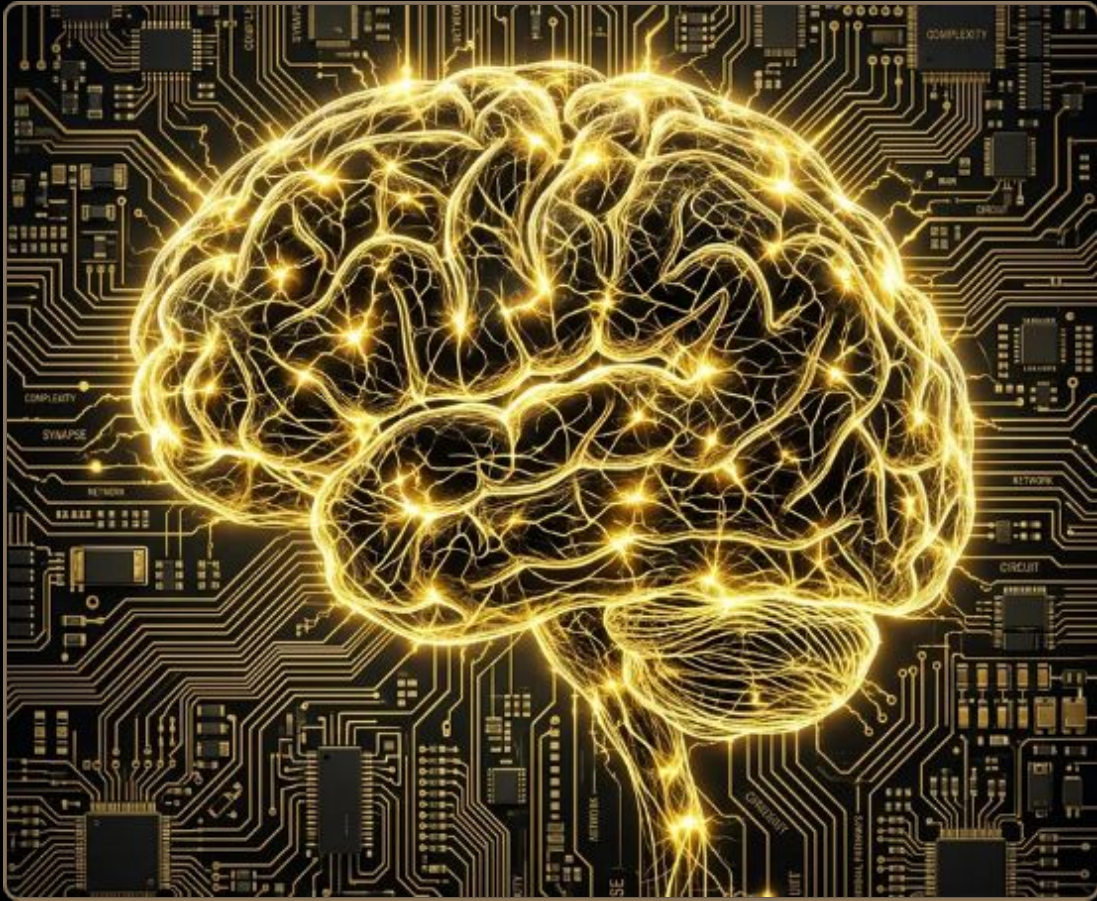
Like a high-end restaurant, MedAssistAI separates its layers for maximum performance and reliability:

- ✓ **Presentation:** React.js & SMS UI.
- ✓ **Application:** Node.js/Express Services.
- ✓ **AI Layer:** Dual-Engine Orchestrator.
- ✓ **Data:** MongoDB & Redis Cache.

Architecture for Data and Application



| AI Engine States & Failover



Sequential Reliability

Ensuring 99.9% uptime for medication reminders:

- 1. Health Check:** Router pings Fadhili AI.
- 2. Failover:** If Fadhili is degraded, requests are instantly routed to Gemini API.
- 3. Recovery:** System monitors Fadhili for recovery to switch back to primary local model.

User Registration Flow

Start: User selects Language (English/Swahili)

→ Validates Input (Phone, Conditions, Emergency Contact)

→ **Security:** System generates PIN & sends via SMS

→ **Verification:** User enters PIN (Max 3 attempts)

→ **Personalization:** Fadhili AI generates custom welcome tip

End: Redirect to Dashboard

| Security & Compliance

Security Layer	Implementation Detail
Authentication	Phone + 6-Digit PIN, JWT Session Tokens
Data Protection	AES-256 Encryption at Rest; TLS 1.3 in Transit
Audit Logging	All PHI access logged for Data Protection Act compliance
Consent	Granular opt-in/opt-out for medical data sharing



| Database Design

The MongoDB Advantage

Flexible JSON-like documents accommodate varied patient needs without rigid schemas:

- ✓ **Collection: Users** (Profiles & Consent)
- ✓ **Collection: Medications** (Schedules)
- ✓ **Collection: Reminders** (Logs & AI Source)
- ✓ **Collection: AI_Logs** (Failover tracking)



Chapter 3 Conclusion

The analysis and design phases establish a secure, localized, and resilient foundation for MedAssistAI, ensuring it meets both the technical and cultural requirements of Kenyan chronic patients.

JKUAT IT Project | 2025

| Image Sources



<https://www.conceptdraw.com/How-To-Guide/picture/Project-Management-Scrum.png>

Source: www.conceptdraw.com



<https://cdn.dribbble.com/userupload/45706491/file/0c6c851f402b9654d69401d588b7dc2a.png?format=webp&resize=400x300&vertical=center>

Source: dribbble.com



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<https://www.slidekit.com/wp-content/uploads/2022/08/Three-Tier-Architecture-Dark-Theme-Slide-jpg.webp>

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Source: easy-peasy.ai



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